Entertainment Data Warehousing(Netflix)

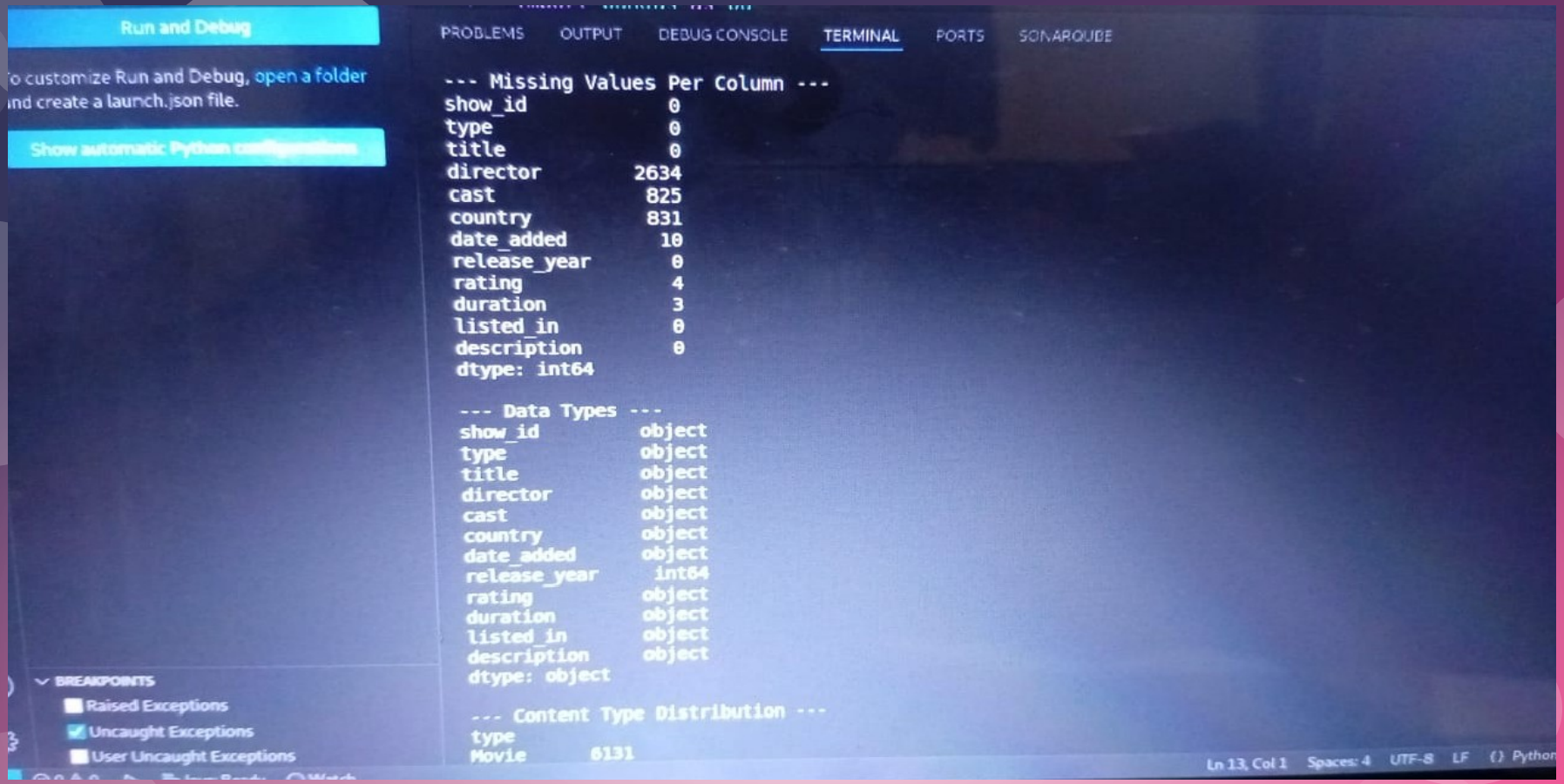
INTRODUCTION

- To design and implement a structured data warehouse that centralizes Netflix's catalog information for efficient querying and analysis.
- Building an ETL pipeline using a global Netflix dataset.
- Goal was to move away from a flat, dirty CSV file and create a structured star schema that is optimized for analytics.

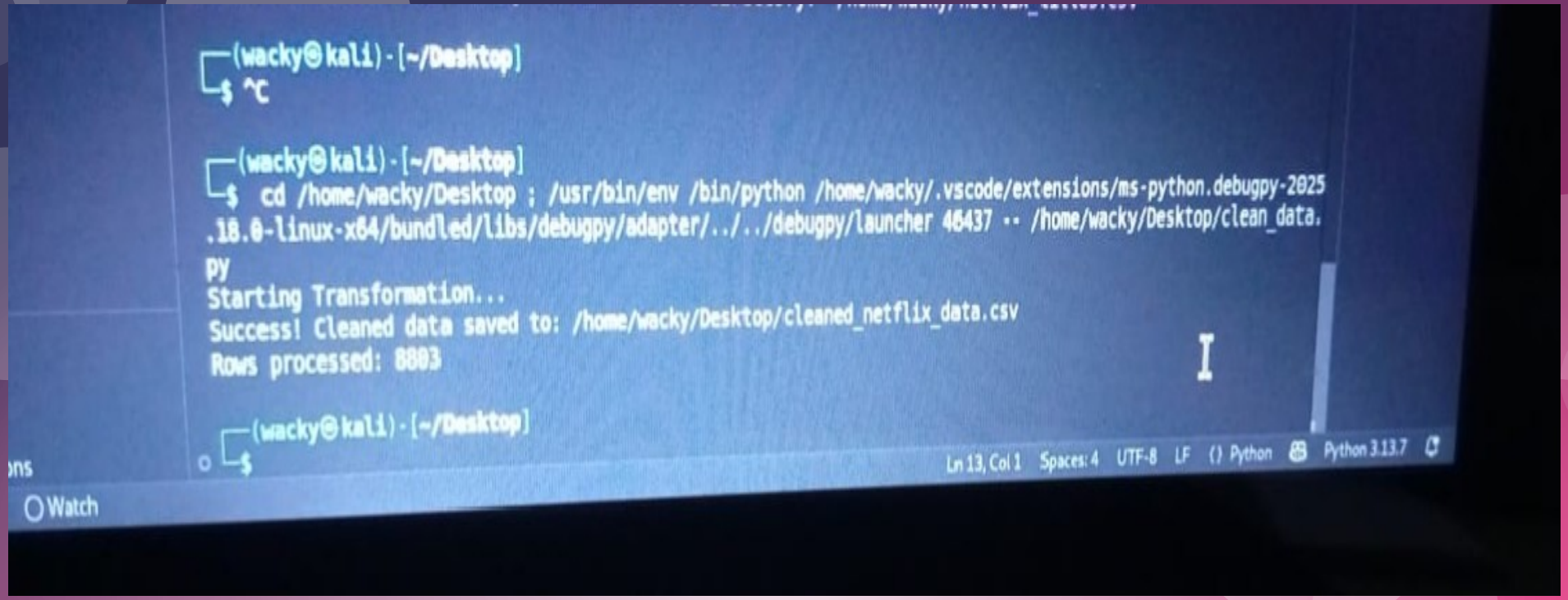
QUICK SUMMARY

- Extracted the raw data directly from the Kaggle API within my Linux environment.
- This gave us access to over 8,800 records, but the data was raw it had many missing values that needed to be fixed before loading.
- In the clean_data.py script, we used Python and Pandas to profile the data.
- Discovered over 2,634 missing entries in the Director column.
- Instead of deleting this data, data imputation was performed, filling those gaps with the label Unknown.
- Standardized the text and removed rows with missing ratings to ensure our warehouse maintains Data Integrity.

Raw data ...


Run and Debug
To customize Run and Debug, open a folder and create a launch.json file.
Show automatic Python configurations
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
SONAROUDE
--- Missing Values Per Column ---
show_id 0
type 0
title 0
director 2634
cast 825
country 831
date_added 10
release_year 0
rating 4
duration 3
listed_in 0
description 0
dtype: int64
--- Data Types ---
show_id object
type object
title object
director object
cast object
country object
date_added object
release_year int64
rating object
duration object
listed_in object
description object
dtype: object
--- Content Type Distribution ---
type
Movie 6131
BREAKPOINTS
Raised Exceptions
Uncaught Exceptions
User Uncaught Exceptions
Ln 13, Col 1
Spaces: 4
UTF-8
LF
Python

Cleaning data...



DIMENSIONAL TABLES

- Dimension tables provide the descriptive context for the data, allowing for categorized analysis.
- dim_type Table: categorizes content into major formats to distinguish between different media types.

  type_id: A unique primary key for each category.

  type_name: The descriptive name of the content type (e.g., "Movie", "TV Show").
- dim_rating Table: standardizes audience maturity ratings across the entire catalog.

  rating_id: A unique identifier for each rating entry.

  rating_code: The specific maturity code (e.g., TV-MA, PG-13, R, TV-14).

DIMENSIONAL TABLES

Creating the Content Type Dimension

```
CREATE TABLE dim_type (
    type_id INTEGER PRIMARY KEY AUTO_INCREMENT,
    type_name TEXT NOT NULL
);
```

- Creating the Content Rating Dimension

```
CREATE TABLE dim_rating (
    rating_id INTEGER PRIMARY KEY AUTO_INCREMENT,
    rating_code TEXT NOT NULL
);
```

FACTS TABLES

- The fact table serves as the central hub, containing the primary records and linking to the dimensions.fact_netflix

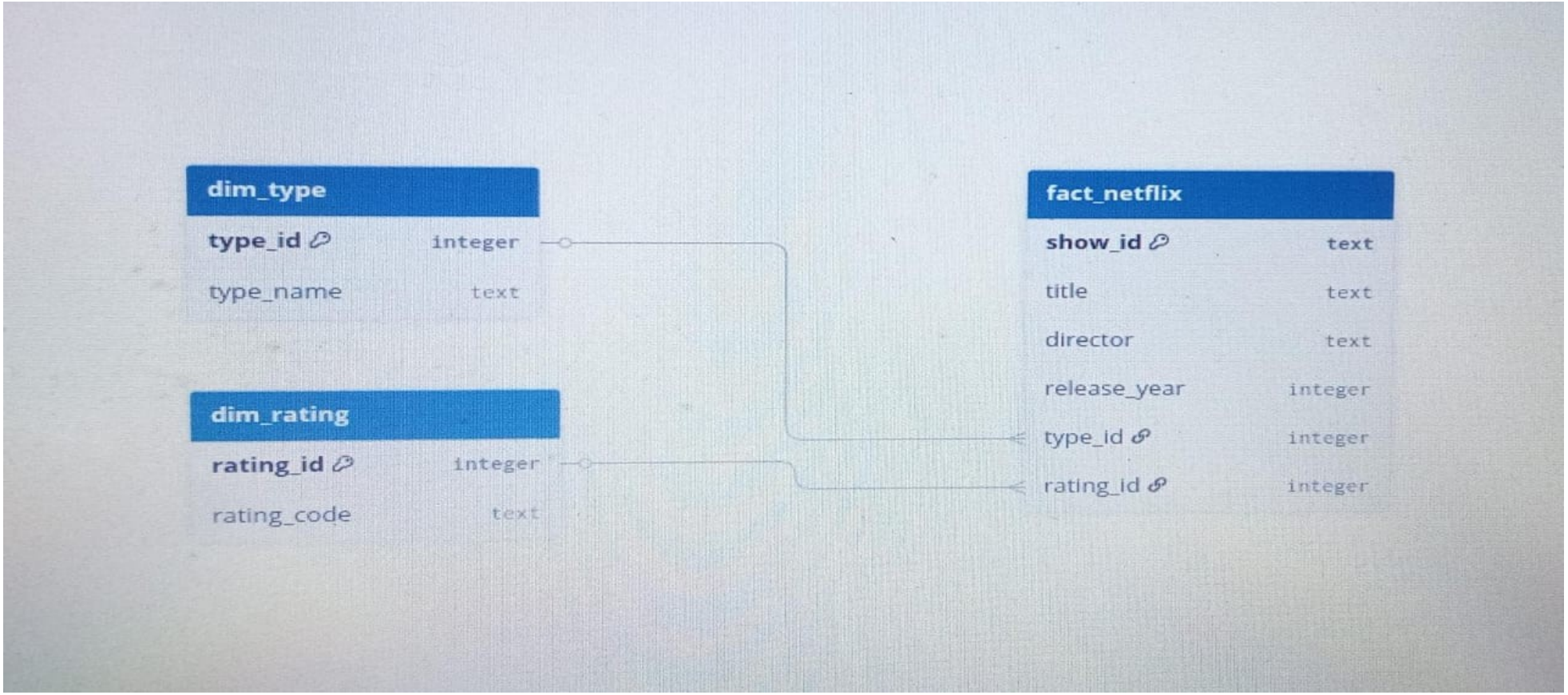
  Table:Primary Key: show_id uniquely identifies each piece of content in the warehouse.

  Core Attributes:title: The official name of the movie or show.director:

- Creating the Central Fact Table

```
CREATE TABLE fact_netflix (
    show_id TEXT PRIMARY KEY,
    title TEXT,
    director TEXT,
    release_year INTEGER,
    type_id INTEGER,
    rating_id INTEGER,
    FOREIGN KEY (type_id) REFERENCES dim_type(type_id),
    FOREIGN KEY (rating_id) REFERENCES dim_rating(rating_id)
);
```

THE DATA (STAR) SCHEMA DIAGRAM



QUERIES

One;

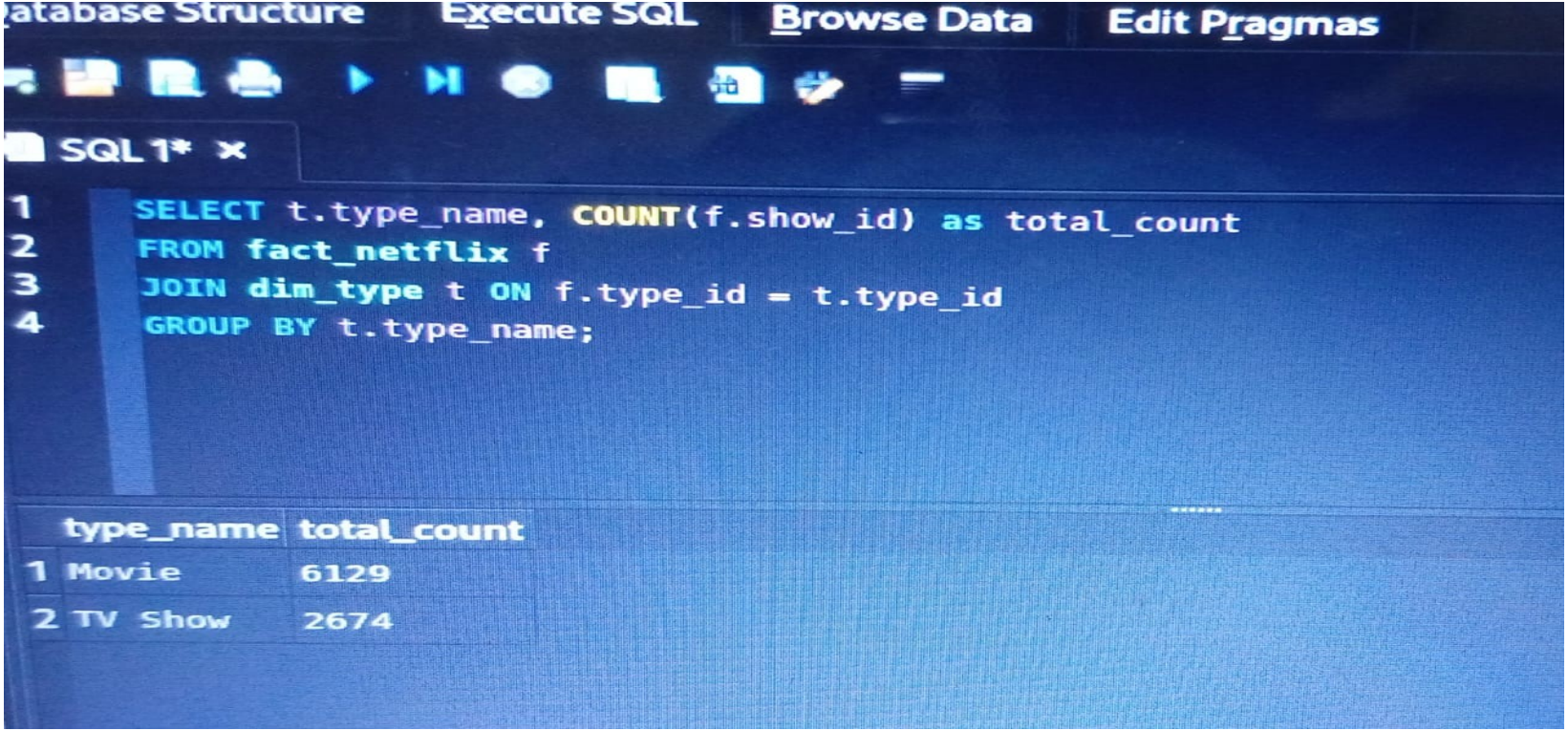
```
SELECT t.type_name, COUNT(f.show_id) as total_count
FROM fact_netflix f
JOIN dim_type t ON f.type_id = t.type_id
GROUP BY t.type_name;
```

Two;

```
SELECT release_year, COUNT(show_id) as release_count
FROM fact_netflix
GROUP BY release_year
ORDER BY release_count DESC
LIMIT 5;
```

SAMPLE FINDINGS

- Content by type distribution;



Continuation ...

SQL1* ×

```sql
SELECT listed_in, COUNT(*) as total_titles
FROM stg_netflix
GROUP BY listed_in
ORDER BY total_titles DESC
LIMIT 10;
```

	listed_in	total_titles
1	Dramas, International Movies	362
2	Documentaries	359
3	Stand-Up Comedy	334
4	Comedies, Dramas, International …	274
5	Dramas, Independent Movies, …	252
6	Kids' TV	220

```
Execution finished without errors.
Result: 10 rows returned in 39ms
At line 1:
SELECT listed_in, COUNT(*) as total_titles
FROM stg_netflix
GROUP BY listed_in
ORDER BY total_titles DESC
LIMIT 10;
```

Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Remote
Identity Select an identity to conn
DBHub.io Local Current Dat
Name
SQL Log Plot DB Schema Remot

Tools Used

- Database:

  SQLite

  netflix_warehouse.db file.
- Language:

  SQL for the DDL and DML.
- IDE:

  VS Code

  DB Browser for SQLite

CONCLUSION

- We have successfully transformed raw, flat Netflix data into a structured Star Schema.
- This architecture reduces data redundancy and significantly improves query performance for large-scale content analysis.